Task	Prompt	Answer
maximal independent set	The task is to determine the maximal independent set guaranteed to contain a given node in the graph. An independent set is a set of nodes such that the subgraph induced by these nodes contains no edges. A maximal independent set is an independent set such that it is not possible to add a new node and still get an independent set. The input graph is guaranteed to be undirected. Here is an undirected graph containing nodes from 1 to 6. The edges are: (1, 2), (1, 6), (1, 3), (2, 3), (2, 4), (2, 5), (3, 5), (4, 5). Question: What is the maximal independent set that includes node 4 of the graph? You need to format your answer as a list of nodes in ascending order, e.g., [node-1, node-2, ..., node-n].	[3, 4, 6]
maximum flow	The task is to determine the value of the maximum flow for the given source node and sink node. The maximum flow is the greatest amount of flow that can be sent from the source to the sink without violating capacity constraints. Here is a directed graph containing nodes from 1 to 5. The edges are: (2, 5, 8), (3, 1, 9), (3, 5, 3), (4, 2, 4). (u ,v, w) denotes the edge from node *u* to node *v* has a capacity of *w*. Question: What is the value of the maximum flow from node 3 to node 2? You need to format your answer as a float number.	0.0
min edge covering	The task is to determine the minimum edge covering of a graph. An edge cover is a set of edges such that every vertex in the graph is incident to at least one edge in the set. The minimum edge cover is the edge cover with the smallest number of edges. The input graph is guaranteed to be undirected. Here is an undirected graph containing nodes from 1 to 9. The edges are: (1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (2, 5), (3, 4), (3, 6), (3, 7), (3, 8), (3, 5), (4, 7), (4, 8), (5, 6), (5, 7), (6, 7), (6, 9), (7, 9). Question: What is the minimum edge covering of the graph? You need to format your answer as a list of edges in ascending dictionary order, e.g., [(u1, v1), (u2, v2), ..., (un, vn)].	[(2, 1), (5, 2), (7, 4), (8, 3), (9, 6)]
min vertex cover	The task is to determine the minimum vertex cover of a graph. A vertex cover is a set of nodes such that every edge in the graph is incident to at least one node in the set. Here is an undirected graph containing nodes from 1 to 5. The edges are: (1, 2), (2, 3), (3, 5), (5, 4). Question: What is the minimum vertex cover of the graph? You need to format your answer as a list of nodes in ascending order, e.g., [node-1, node-2, ..., node-n].	[2, 5]
minimum spanning tree	The task is to determine the minimum spanning tree of a graph. A minimum spanning tree is a subset of the edges that connects all vertices in the graph with the minimum possible total edge weight. If not specified, all edges have equal edge weights. The input graph is guaranteed to be undirected and connected. Here is an undirected graph containing nodes from 1 to 9. The edges are: (1, 2), (1, 8), (1, 5), (1, 6), (1, 4), (1, 7), (1, 9), (2, 5), (2, 6), (2, 4), (2, 7), (2, 3), (8, 3), (8, 4), (8, 6), (8, 7), (5, 3), (5, 4), (5, 6), (5, 7), (5, 9), (6, 3), (6, 4), (6, 7), (6, 9), (4, 3), (4, 7), (4, 9), (7, 9), (9, 3). Question: What is the minimum spanning tree of the graph? You need to format your answer as a list of edges in ascending dictionary order, e.g., [(u1, v1), (u2, v2), ..., (un, vn)].	[(1, 2), (1, 4), (1, 5), (1, 6), (1, 7), (1, 8), (1, 9), (2, 3)]